Quadrilaterals

Geometry · Edexcel IGCSE Mathematics A (4MA1) · Year 10 · worksheet

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Name	Class	Date

Answer ALL questions.
Write your answers in the spaces provided. You must write down all the stages in your working.
Suggested time: 88 minutes · Total marks: 88

KEY RULES

Angle sum	The interior angles of any quadrilateral add to 360°
Parallelogram	Opposite sides equal and parallel; opposite angles equal; adjacent angles add to 180°
Rhombus	All sides equal; diagonals bisect each other at 90°
Kite	Two pairs of equal adjacent sides; one pair of equal opposite angles
Trapezium	One pair of parallel sides; area $= \frac{1}{2}(a+b)h$
Rectangle / square	All angles 90°; a square also has four equal sides

WORKED EXAMPLE 1

[2 marks]

$ABCD$ is a parallelogram. Angle $A = 65^\circ$. Work out angles B and C.

Step 1	Adjacent angles add to 180	$B = 180 - 65 = 115^\circ$
Step 2	Opposite angles are equal	$C = A = 65^\circ$

Answer: $B = 115^\circ,\ C = 65^\circ$

WORKED EXAMPLE 2

[3 marks]

A kite has angles 110°, 50°, x° and x°. Work out x.

Step 1	The angles of a quadrilateral add to 360	$110 + 50 + x + x = 360$
Step 2	Simplify	$2x = 200$
Step 3	Divide by 2	$x = 100$

Answer: $x = 100$

WORKED EXAMPLE 3

[3 marks]

A trapezium has parallel sides 7 cm and 11 cm, and height 5 cm. Work out its area.

Step 1	Area = $\frac{1}{2}(a+b)h$	$\frac{1}{2}(7+11) \times 5$
Step 2	Work out	$9 \times 5 = 45$

Answer: $45\ \text{cm}^2$

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Quadrilaterals

1 A quadrilateral has diagonals that are equal in length, bisect each other and cross

What is the quadrilateral?

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

2 Opposite angles of a parallelogram are $(3x + 10)^\circ$ and $(5x - 20)^\circ$.

Work out x.

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

3 **13** The diagram shows a trapezium.

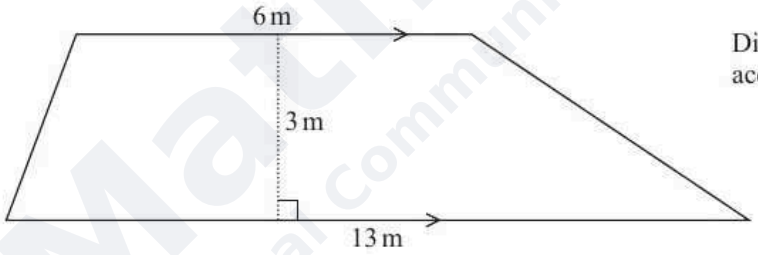


Diagram **NOT** accurately drawn

Work out the area of the trapezium.

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4 **15** Here is a trapezium.

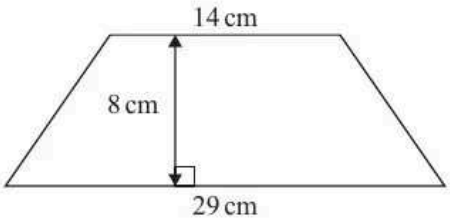


Diagram **NOT** accurately drawn

Work out the area of the trapezium.

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5 $ABCD$ is a rhombus. Angle $ABC = (2x + 10)^\circ$ and angle $BCD = (3x - 20)^\circ$.

Work out the size of angle ABC.

(Total for question = 3 marks)

Practice question written for this worksheet (not from a past paper)

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6 The diagram shows a kite $ABCD$

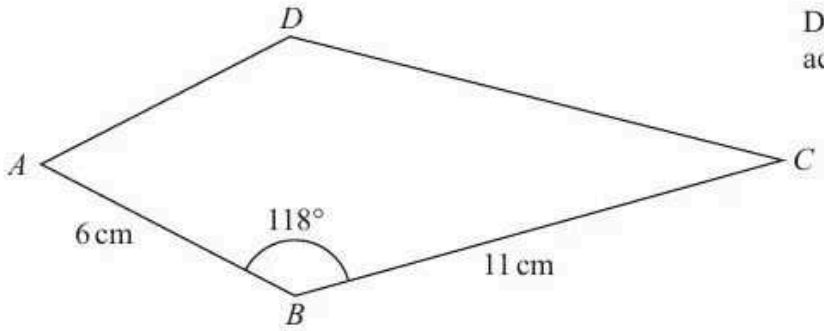


Diagram **NOT** accurately drawn

$AB = 6\,\text{cm}$ $BC = 11\,\text{cm}$ Angle $ABC = 118°$

Calculate the area of the kite.
Give your answer correct to 3 significant figures.

..... cm^2

(Total for Question 15 is 3 marks)

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7 **15**

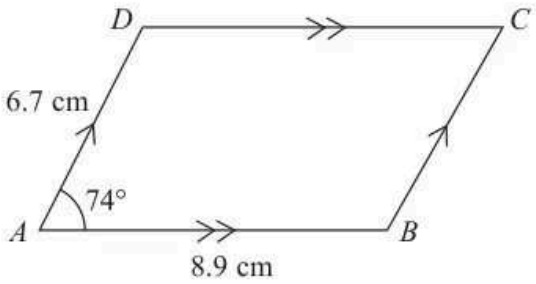


Diagram **NOT** accurately drawn

ABCD is a parallelogram.
AB = 8.9 cm.
AD = 6.7 cm.
Angle *BAD* = 74°

Calculate the area of parallelogram *ABCD*.
Give your answer correct to 3 significant figures.

..... cm²

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8 *ABCD* is a kite.

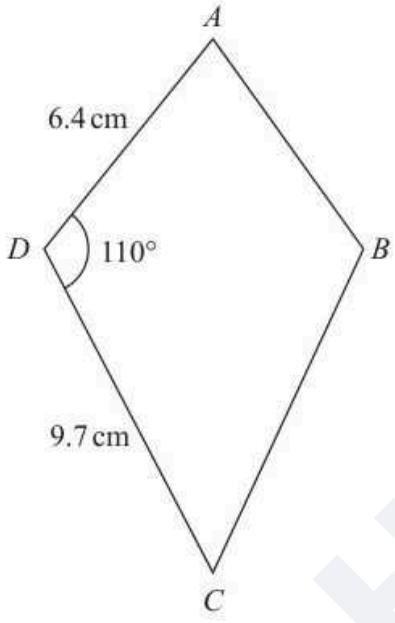


Diagram **NOT** accurately drawn

Work out the area of the kite.
Give your answer correct to 3 significant figures.

..... cm²

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9 7

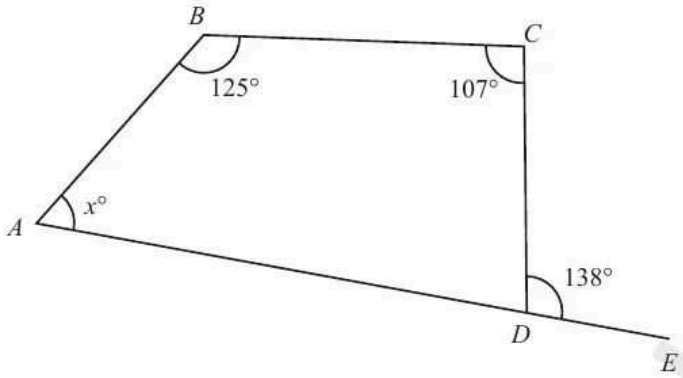


Diagram **NOT** accurately drawn

ABCD is a quadrilateral.

ADE is a straight line.

Work out the value of x
Give a reason for each stage of your working.

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10 9 Here is a quadrilateral.

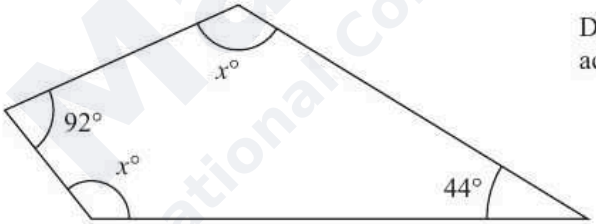


Diagram **NOT** accurately drawn

(i) Work out the value of x

(3)

(ii) Give a reason for your answer.

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11 **13** The diagram shows a classroom wall in the shape of a trapezium.

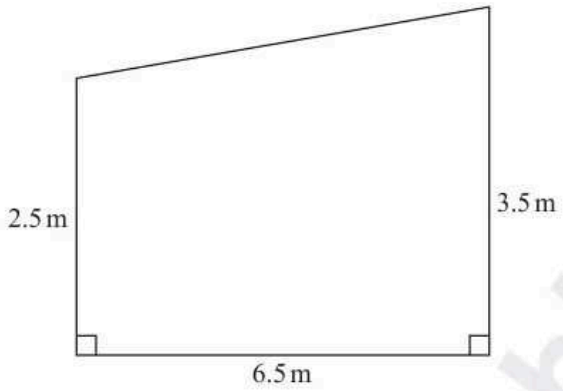


Diagram **NOT** accurately drawn

Dion wants to paint the classroom wall completely twice.
He knows that each tin of paint will cover $12\,m^2$

He is going to have to buy all the paint he needs.

Work out the least number of tins of paint that Dion will need to buy.
Show your working clearly.

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12 $ABCD$ is a trapezium.

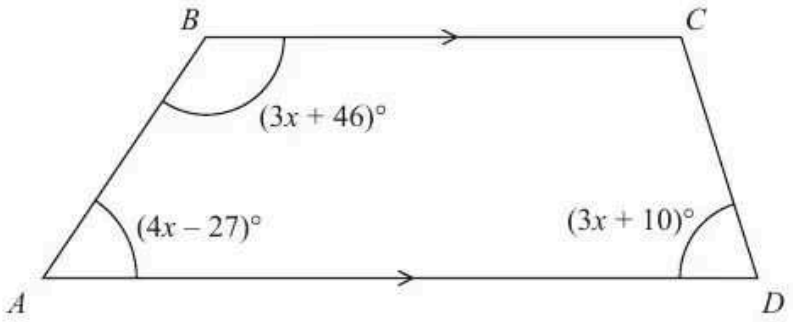


Diagram **NOT** accurately drawn

BC is parallel to AD

Find the size of the largest angle inside the trapezium.

.....

(Total for Question 3 is 4 marks)

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13 The diagram shows rectangle $ABCD$

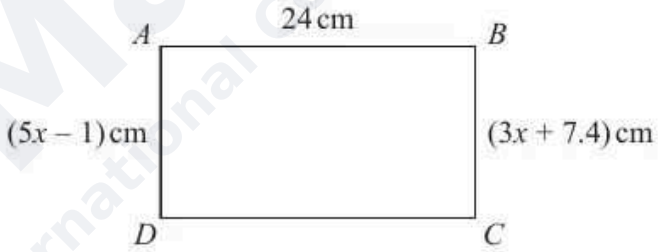


Diagram **NOT** accurately drawn

Work out the perimeter of the rectangle.
Show your working clearly.

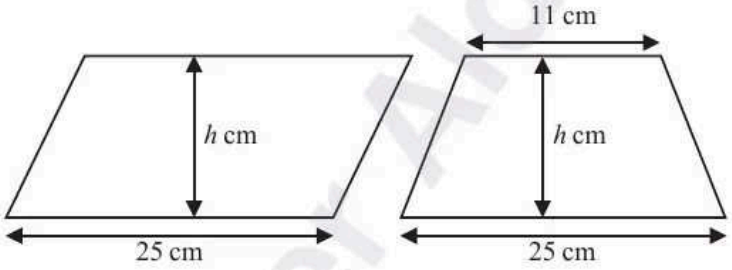
..... cm

(Total for Question 4 is 4 marks)

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14 **17** The diagram shows a parallelogram and a trapezium.

Diagram **NOT** accurately drawn



Both shapes have a base of 25 cm and a perpendicular height of h cm

The area of the parallelogram is 175 cm^2

Work out the area of the trapezium as a percentage of the area of the parallelogram.

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15 3

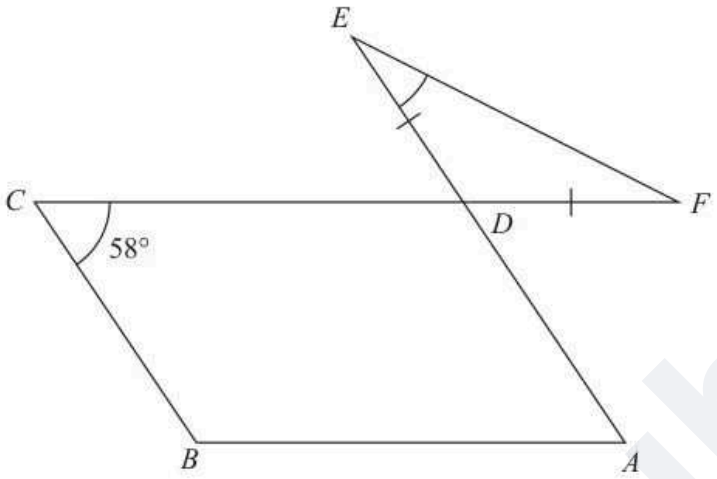


Diagram **NOT** accurately drawn

The diagram shows a parallelogram $ABCD$ and an isosceles triangle DEF in which $DE = DF$

CDF and ADE are straight lines.
Angle $BCD = 58°$

Work out the size of angle DEF.
Give a reason for each stage of your working.

.....°

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16 The diagram shows a trapezium $PQRS$.

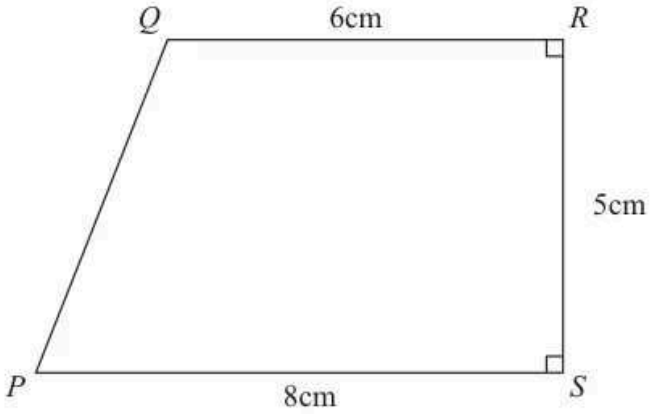


Diagram **NOT** accurately drawn

(a) Calculate the area of the trapezium $PQRS$.

..... cm^2

(2)

(b) Calculate the length PQ.
Give your answer correct to 3 significant figures.

..... cm

(4)

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17 **8** The diagram shows a trapezium, $ABCD$

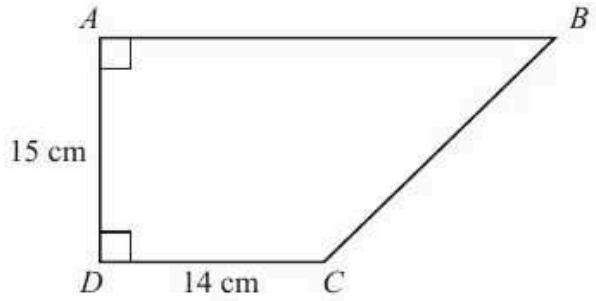


Diagram **NOT** accurately drawn

DAB and ADC are right angles.

$AD = 15$ cm $DC = 14$ cm

The area of the trapezium is 360 cm²

Work out the perimeter of the trapezium.

..... cm

(Total for Question 8 is 6 marks)

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CHALLENGE QUESTIONS

18 9

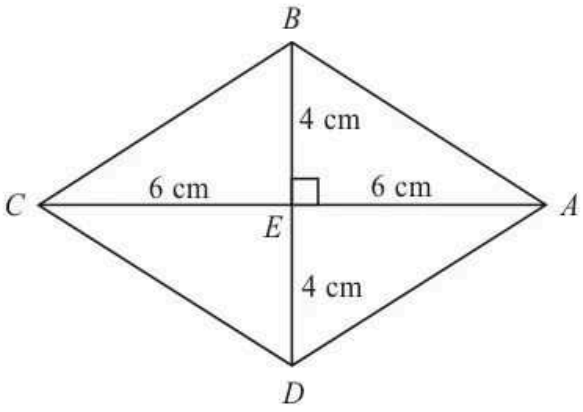


Diagram **NOT** accurately drawn

ABCD is a rhombus.
The diagonals *AC* and *BD* cross at the point *E*.
AE = *CE* = 6 cm.
BE = *DE* = 4 cm.
Angle *AEB* = 90°

(a) Work out the area of the rhombus.

..... cm²
(3)

(b) Work out the length of *AB*.
Give your answer correct to 3 significant figures.

..... cm
(3)

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19 9

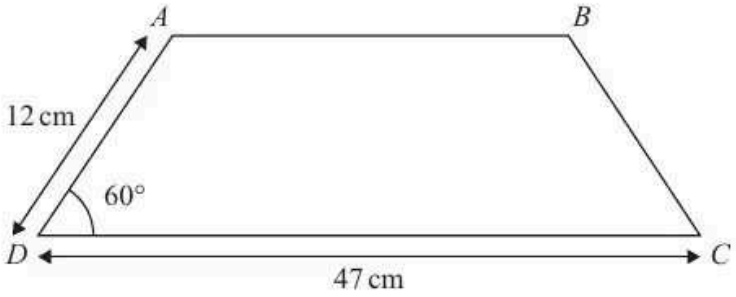


Diagram **NOT** accurately drawn

ABCD is a trapezium with one line of symmetry.

angle $ADC = 60°$ $AD = 12\,\text{cm}$ $DC = 47\,\text{cm}$

Work out the area of the trapezium.
Give your answer correct to 3 significant figures.
Show your working clearly.

..... cm^2

(Total for Question 9 is 5 marks)

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20 The diagram shows parallelogram *EFGH*.

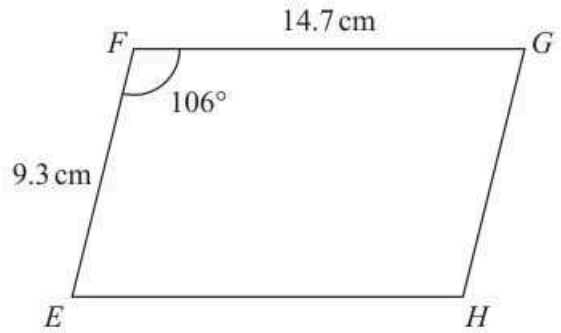


Diagram **NOT** accurately drawn

$EF = 9.3$ cm
$FG = 14.7$ cm
Angle $EFG = 106°$

(a) Work out the area of the parallelogram.
Give your answer correct to 3 significant figures.

..... cm^2
(2)

(b) Work out the length of the diagonal EG of the parallelogram.
Give your answer correct to 3 significant figures.

..... cm
(3)

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21 Here is a parallelogram $PQRS$, in which angle SPQ is acute.

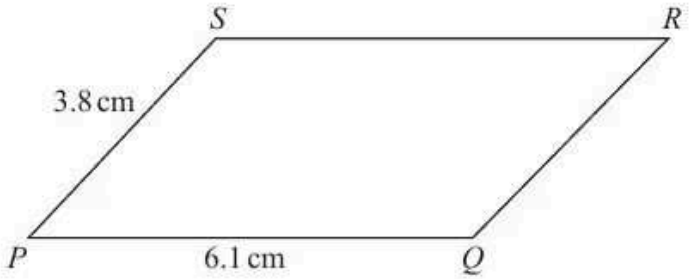


Diagram **NOT** accurately drawn

$PQ = 6.1\,\text{cm}$ $PS = 3.8\,\text{cm}$

The area of the parallelogram is $18\,\text{cm}^2$

Work out the length of QS
Give your answer correct to 3 significant figures.

..... cm

(Total for Question 17 is 5 marks)

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22 The diagram shows trapezium $ABCD$ in which BC and AD are parallel.

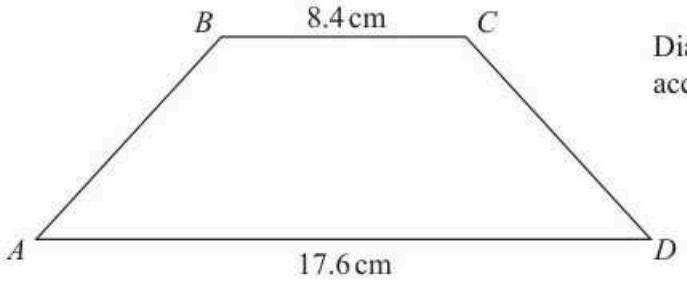


Diagram **NOT** accurately drawn

The trapezium has exactly one line of symmetry.

$BC = 8.4\,\text{cm}$
$AD = 17.6\,\text{cm}$

The trapezium has area $179.4\,\text{cm}^2$

Work out the size of angle ABC.
Give your answer correct to 1 decimal place.

.....°

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TOTAL FOR PAPER IS 88 MARKS